

Database Design and Development (CS4092) – Summer 2026

Final Project – Library Management System

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Phase 1: Requirements Gathering

Introduction

For my final project, I am building a backend database for a library. I chose to adapt the sample e-commerce project into a library domain because it maps well onto the same relationship while giving me a system that is easier to reason about from personal experience. This document covers the first phase of the project, which is requirements gathering.

System Overview

The system supports a library where members can browse and borrow books, and staff manage the book collection and track loans. There are two main user roles in this system, staff and members. Each role has a different set of permissions and use cases, which I outline below.

User Roles and Use Cases

Staff

Staff members are responsible for managing the book catalog and borrowing activity. Their use cases include the following.

- Add a new book to the catalog, including title, author, and available copies.
- Edit an existing book record, such as updating the number of available copies.
- View all books in the catalog, and filter by availability or category if needed.
- View a member's loan history to check for overdue books or outstanding fines.
- Record when a book is returned and calculate a fine if the return is late.

Member

Members use the system to find and borrow books. Their use cases include the following.

- Register for a library account with basic contact information.
- Browse the catalog to see which books are available.
- Borrow a book, which creates a loan record.
- View their own current and past loans, including due dates.
- View any fines they owe for late returns.

Data Requirements

Member

A member needs a *unique ID, full name, email address, phone number, and the date* they joined the library. The email should be unique so that each person only has one account.

Book

A book needs a *unique ID, title, author, category, total copies owned by the library, and copies currently available*. I am separating total copies from available copies so the system can track how many are checked out at any given time without recalculating from loan records every time.

Staff

A staff member needs a *unique ID, full name, email address, job title, and hire date*.

Loan

A loan represents one borrowing transaction. It needs a *unique ID, the member ID who borrowed the book, the book ID being borrowed, the staff ID who processed the checkout, the date borrowed, the due date, and the date returned*, which can be empty if the book has not been returned yet.

Fine

A fine represents a late fee charged to a member for an overdue return. It needs a *unique ID, the loan ID it is associated with, the amount owed, whether it has been paid, and the date it was paid if applicable*.

Key Assumptions

I want to state a few assumptions clearly, since they affect how I design the schema in the next phase.

A member can only borrow one copy of the same book at a time. This keeps the individual project scope manageable while still meeting the minimum table requirement.

A loan is considered active until a return date is recorded. Once a book is marked as returned, the available copy count for that book increases by one.

Fines are only generated when a book is returned late. I am not building in a real payment processing system, since that is outside the scope of this course project.

Next Steps

The next phase of this project is the entity relationship diagram, where I will translate these entities and their relationships into a visual model with clearly defined keys and connections before moving into schema design.